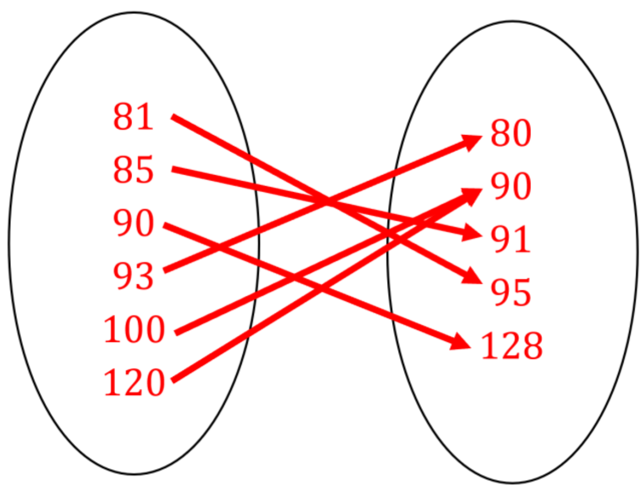
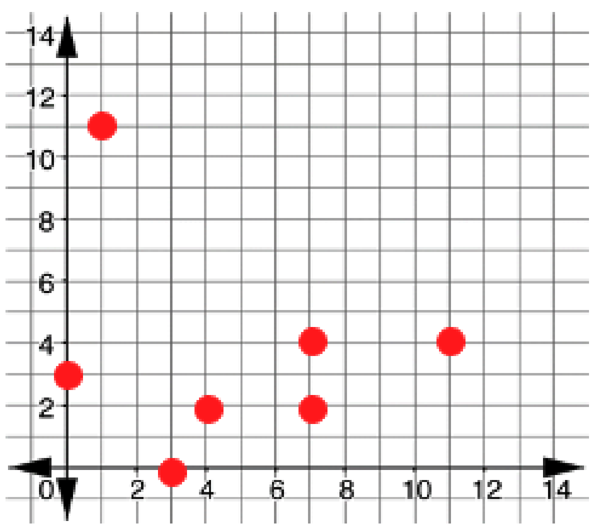
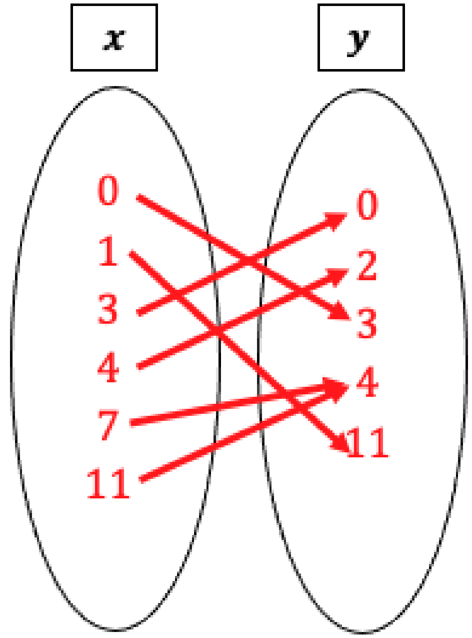
8th Grade Unit 10 Assessment Guide

General Information	When assessing these benchmarks, it is easy to create questions where students can easily guess an answer. If editing this assessment, keep in mind what information from student responses would provide you with a complete picture of a student's understanding while taking into account the possibility of a student guessing an answer. Just stating whether a relation is a function or not is an example of a type of question that can easily be answered correctly without student understanding. In the same regard, be careful about how to distribute partial credit.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	**1a**	**Benchmark(s)**	MA.8.F.1.1
		**Recommended Scoring** **(12 Total Points)**	Students do not have to order the values in the mapping diagram. Suggested points are **3** points for the inputs, **3** points for the outputs, and **6** points for the correct mapping (one point for each mapping).

Question	1b	Benchmark(s)	MA.8.F.1.1
The relationship between the maximum descent angle and maximum speed of the roller coasters ● is ○ is not a function. The domain of the relationship is {81, 85, 90, 93, 100, 120}.		Recommended Scoring (6 Total Points)	Students should not get partial credit for the box. Consider partial credit if a student forgets a number in the domain. Suggested points is **6** points.
Question	**2**	**Benchmark(s)**	MA.8.F.1.2
○ Only relation K could represent a ● Only relation R could represent a ○ Both relation K and R could represent ○ Neither relation K and R can represent a		**Recommended Scoring** **(4 Total Points)**	Suggested points is **4** points.
Question	**3a**	**Benchmark(s)**	MA.8.F.1.1
<table><tr><th>Domain</th><th>Range</th></tr><tr><td>0, 1, 3, 4, 7, 11</td><td>0, 2, 3, 4, 11</td></tr></table>		**Recommended Scoring** **(12 Total Points)**	The domain and range can be considered as separate responses. Suggested points for each is **6** points.
		Additional Notes	Students can repeat values in the domain and/or range.

<table>
<tr><th>Question</th><th>3b</th><th>Benchmark(s)</th><td>MA.8.F.1.1</td></tr>
<tr><td colspan="2" rowspan="2">
<table>
<tr><th>Graph</th><th>Mapping Diagram</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>
</td><th>Recommended Scoring

(17 Total Points)</th><td>Suggested points is 7 points for the graph (1 point for each correctly graphed point). Suggested points for the mapping diagram is 3 points for the inputs, 3 points for the outputs, and 6 points for the correct mapping (one for each correct arrow).</td></tr>
<tr><th>Additional Notes</th><td>Consider deducting 1 point for minor errors. If students make more than one minor error, then consider deducting no more than 2 points.</td></tr>
<tr><th>Question</th><th>3c</th><th>Benchmark(s)</th><td>MA.8.F.1.1</td></tr>
<tr><td colspan="2">If the ordered pair __(7,2) or (7,4)__ is removed, then the relation is a function.</td><th>Recommended Scoring

(4 Total Points)</th><td>Suggested points for either ordered pair is 4 points.</td></tr>
</table>

Question	**4a**	**Benchmark(s)**	MA.8.F.1.2
Group A has ○ 0 ○ 1 ○ 2 ● 3 ○ 4 equations whose graph would be a line. Group B has ○ 0 ○ 1 ● 2 ○ 3 ○ 4 equations whose graph would be a line.		**Recommended Scoring** **(8 Total Points)**	Each sentence should be considered separate responses. Suggested points for each sentence is **4** points.
		Additional Notes	Students may make notes on the table to indicate which equations are linear. Consider partial credit if a student identifies some, but not all of the linear equations in a given group (up to 1 point for each group).

Question	**4b**	**Benchmark(s)**	MA.8.F.1.2
$y = \frac{4}{5}x^2 + 1$		**Recommended Scoring** **(4 Total Points)**	Suggested points for is **4** points.

Question	5a	Benchmark(s)	MA.8.F.1.3
Range	$0 \leq y \leq 20$	**Recommended Scoring** **(8 Total Points)**	Suggested total points is **7** points, **3** points each for the correct values of the range and **2** points for the use of y.
		Additional Notes	A common error is students use x instead of y. This is often because the variable x is a commonly used variable. Students could also correctly write the answer as two separate inequalities, $y \geq 0$ and $y \leq 20$ for full credit.

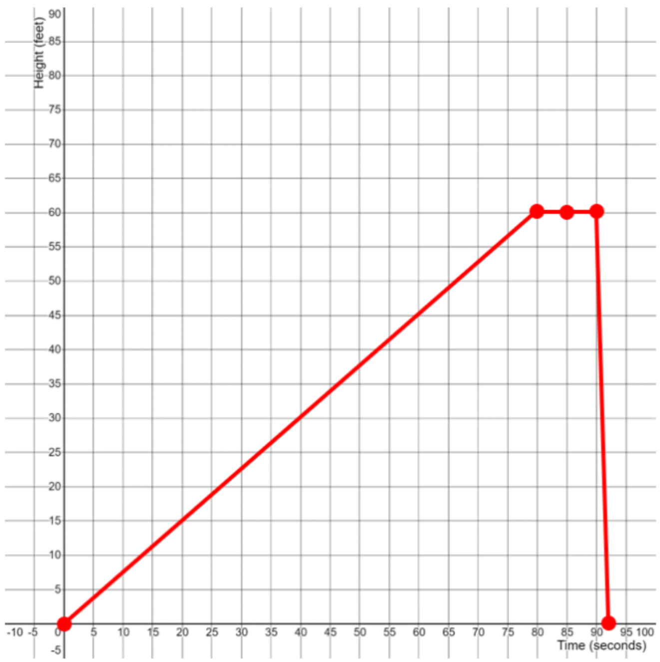
<table>
<tr><th>Question</th><th>5b</th><th>Benchmark(s)</th><td>MA.8.F.1.3</td></tr>
<tr><td colspan="2" rowspan="2">

Decreasing	$4 \leq x \leq 5$

</td><th>Recommended Scoring

(8 Total Points)</th><td>Suggested total points is 8 points, 3 points each for the correct values of the domain and 2 points for the use of x.</td></tr>
<tr><th>Additional Notes</th><td>Students may respond with $6.5 \leq y \leq 7.5$. Consider the information this provides about the student. While they did not correctly answer the question, they are indicating the understand which part of the graph shows a decrease.</td></tr>
<tr><th>Question</th><th>5c</th><th>Benchmark(s)</th><td>MA.8.F.1.3</td></tr>
<tr><td colspan="2" rowspan="2">The surface area remains constant.</td><th>Recommended Scoring

(4 Total Points)</th><td>Suggested points for is 4 points.</td></tr>
<tr><th>Additional Notes</th><td>Students may word their response differently. Consider other correct descriptions.</td></tr>
</table>

<table>
<tr><th>Question</th><th>6a</th><th>Benchmark(s)</th><td>MA.8.F.1.3</td></tr>
<tr><td colspan="2" rowspan="2">
</td><th>Recommended Scoring

(9 Total Points)</th><td>Suggested points is 3 points for each piece;
• Increasing
• Constant
• Decreasing</td></tr>
<tr><th>Additional Notes</th><td>Students do not have to mark the value at $(85,60)$.
Consider awarding 1 of the 3 points for the constant piece if a student only includes a constant height of 60 feet for 5 seconds then draws the decreasing piece. See note on question 6b if a student continues this error.</td></tr>
<tr><th>Question</th><th>6b</th><th>Benchmark(s)</th><td>MA.8.F.1.3</td></tr>
<tr><td colspan="2" rowspan="2">Isabella goes the fastest between 90 and 92 seconds.</td><th>Recommended Scoring

(4 Total Points)</th><td>Suggested points is 4 points for the correct interval.</td></tr>
<tr><th>Additional Notes</th><td>If students make the error described in part 6a, they may say she goes fastest between 85 and 87 seconds. Consider no further deduction because the student correctly answered this question based on their mistake.</td></tr>
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